

Trickery with LD Preload

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Project idea:

The goal of this project is to develop a shared library that intercepts commonly used file, network, and memory-related functions.

We want to load our own shared library before any program loads their designated libraries using LD_Preload. This way our program uses the functions of our own shared libraries instead the standard C libraries. We want to use strace to see what commands a function uses before trying to hijack them.

The project will be implemented in C on a Linux system. We have not yet fully decided which functions we would like to hijack but we take the following functions into consideration:

- **fopen:** When opening a file, a restriction will stop the user and is possibly asked for a password.
- **fclose:** When closing a file, the action gets denied and the user cannot close the file.
- **read:** When reading a file, fake file content is being shown.
- **write:** When writing in a file, fake file content is being written.
- **print:** When trying to print something, our shared library will print some fun joke.
- **remove:** When trying to remove an important file (PDF), the user will receive a warning and the file will not be deleted.
- **rename:** When trying to rename a file (PDF) or a folder, the user will receive a warning and the file/folder will not be renamed.
- **malloc/free:** We can track where and how much memory is used and where memory was never freed.
- **strcmp:** Every string that is typed in as a password gets passed as correct.
- **connect:** When connecting to a specific network (adam.unibas.ch), a restriction will stop the user.

Requirements:

- Linux OS
- GCC compiler to be able to compile the C code.
- Possibly we will use ncurses to implement a simple terminal interface.

Risks:

- Recursion, in case that functions call other functions internally.
- Handling functions with variable arguments.